Image Recognition with Deep Learning

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***Abstract*—Image recognition is one of the most important fields of image processing and computer vision. Food image classification is an unique branch of image recognition problem. In modern days people are more conscious about their health. A system that can classify food from image is necessary for a dietary assessment system. Classification of food images is very challenging since the dataset of food images is highly non-linear. In this paper we proposed a method that can classify food categories with images. We used convolutional neural network to classify food images. The CNNs are a very effective class of neural networks that is highly effective at the task of image classifying, object detection and other computer vision problems. We classified a food dataset consisting different food categories with 16643 images. We obtained an accuracy of 92.86% in our experiment.**

Keywords—Deep Learning, CNN, Computer Vision, Image Processin

I. Introduction

In the present day obesity and other health problems are rising day by day. According to [1] obesity doubled since 1980 in more than 70 countries. The obesity may lead to other types of chronic disease such as heart disease, diabetes, arthritis, etc. There should be given more importance to the nutritional value of food to prevent these diseases. Dietary management is the key to regularize the food habits of people. It will help people with dietary management if they know the information about the food they are eating. To get information about the food a system is required that will detect the food from image and then analyze the dietary and calorie information. In this paper the researchers mainly concerned with the detection of food from images.

Image processing and computer vision techniques are now being applied in many domains. One of the domains is food recognition from image. Food recognition is a challenging task since there is large similarity between food classes. Besides any image recognition also needs more computation power than most of the text base data classification. But to benefit from food recognition model people should be able use it in a less expensive device. In present day there are affordable smartphones with high computation power than can process high quality image data. So the model described in this paper can be implemented in smartphones.

Food classification from image is a challenging task because same class of foods may have many differences in their image. In [2] they implemented k-nearest neighbor algorithm and vocabulary tree algorithm to classify 42 food categories with 1453 images. To measure the distance they chose L1-norm for the Stochastic Coordinate Descent (SCD), EFD and GFD features and the Euclidean distance (L2-norm) for the Dual Coordinate Descent (DCD) feature. They got 84.2 % top 4 accuracy and 64.5 % top 1 accuracy with combinations of DCD, MDSIFT, SCD, SIFT features.

In [3] they proposed a method that applies SIFT and LBP (Local Binary Pattern) features in SVM classifier with PFI dataset. SIFT feature is used detect and describe local features in images and LBP is a type of visual descriptor and was used because it has many advantages. LBP is simple to compute, immune to illumination changes.

In [4] they proposed a method that classifies food images with sphere shaped support vector machine. The support vector machine can efficiently perform non-linear classification with kernel trick. They applied this method to FoodLog dataset which consisting of 6512 images. They used FCM algorithm to classify food images. FCM is similar to k-means clustering algorithm. In FCM at first coefficients are assigned randomly to each data point for being in the clusters then centroid for each cluster is computed and for each data point coefficients is computed of being in the clusters and until convergence these steps are

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repeated. After applying FCM to segment food images they used sphere shaped support vector machine (SVM) to classify segmented images. In this experiment they got an accuracy of 95%.

The authors of [5] used random forests classifier to classify food-101 dataset. Food-101 is a large dataset with 101 categories. The researchers got 50.76% of accuracy with the RFDC approach.

In recent literatures there are also several approaches that used deep convolution neural network to classify food images. Since convolutional neural networks are scalable for large datasets it is more suitable to use CNNs for food image classification. Deep learning was used in [6] to classify UEC-256 food image for dietary assessment system using digital device. They obtained a top-1 accuracy of 54.7 % and 81.5% accuracy of top-5.

A Pre-trained DNN was applied in [7]. The deep CNN was pre-trained on ImageNet with 1000 food related categories than fine tuned to classify UEC-FOOD100 dataset. They achieved 78.77% top-1 accuracy.

In [8] they used GoogLeNet to classify Thai fast food images in TFF food dataset. They achieved 88.33% accuracy for 11 classes. In [9] they implemented and compared several convolutional neural network models with food-11 dataset. They got 70.12% with their proposed approach, 80.51% with Caffenet and 82.07% accuracy with Alexnet.

In our paper we implemented deep learning with convolution neural network to classify food-11 dataset and we got an accuracy of 92.86%.

II. Methodology

We used convolutional neural network in our approach to classify food images. The convolutional neural network is a category of neural network that has been proven very efficient in image classification. The convolutional neural network learns the filters that in traditional algorithms were hand-engineered. In our method we used an inception v3 [10] model pre-trained with ImageNet [11]. The method of our task classifying food images consisting of four processes:

- Select a food image dataset
- Image Pre-processing
- Train dataset using deep learning algorithm
- Classification of food images

The figure 1 depicts the methodology of our objective.

A. Input Image

We used the Food-11 dataset for our research. The dataset was created by the authors who proposed [7]. The dataset consists of 16643 images grouped into 11 major food categories. The 11 food categories are Bread, Dairy products, Dessert, Egg, Fried food, Meat, Pasta, Rice, Seafood, Soup, and Vegetables. The food images are divided into three parts: training set with 9866 images, validation set with 3430 images and evaluation set with 3347 images.

B. Image Pre-processing

We applied some image pre-processing technique to increase efficiency to our system by speeding up training time and also . The pre-processing such as random rotation and horizontal flips help convolutional neural network models to be insensitive of the exact position of the object in the images. The ZCA whitening reduces the redundancy in the matrix of pixel images and highlights the structures and features of the images to the convolutional neural network. First we resized all our images to 299 x 299 x 3 to increase processing time and also to fit in Inception V3. After that we applied following image pre-processing techniques.

- We set input mean to 0 over the dataset.
- We also set each sample mean to 0.
- Then we divided inputs by standard deviation of the dataset.
- We also ensured that each input is divided its standard deviation.
- We applied ZCA whitening.
- We randomly rotated images in the range from 0 to 180 degrees to make our learning model invariant to the object location in the image.
- Randomly shift images horizontally.
- Randomly shift images vertically.
- Randomly flip images.

C. Deep Learning

Our research utilizes the inception V3[10] model. The architecture of the model is given in figure 2.

The layers of the inception V3 model are:

- Convolution Layer: At the beginning convolution layer with input size 299 x 299 x 3 to create feature maps by convolving input images.
- Max Pooling Layer: Max-pooling is a sample-based discretization process. Max pooling is done by applying a max filter to non-overlapping sub regions of the input matrices. Max-pooling extracts the most important features like vertical edgesand horizontal edges [12].

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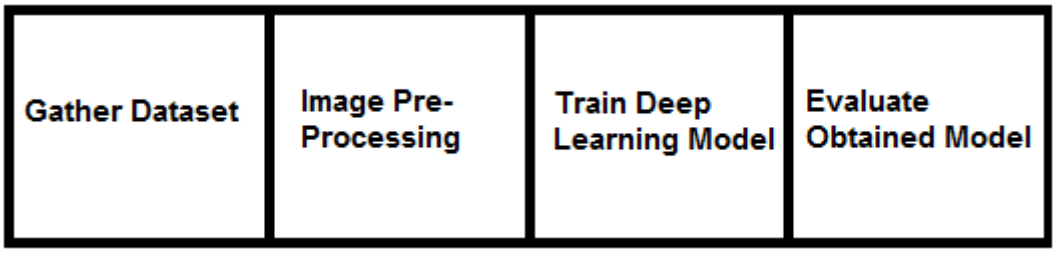

Fig. 1. Methodology of food image classification

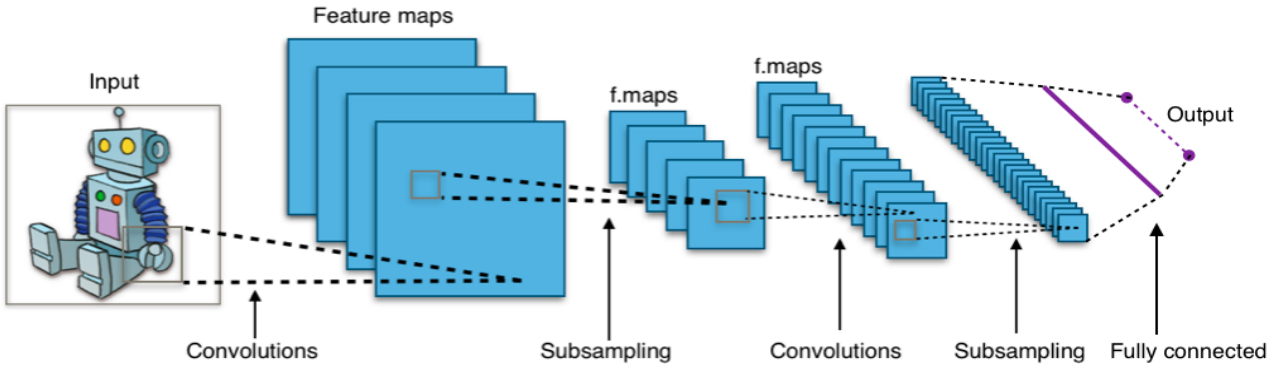

Fig. 2. Typical CNN Architecture.

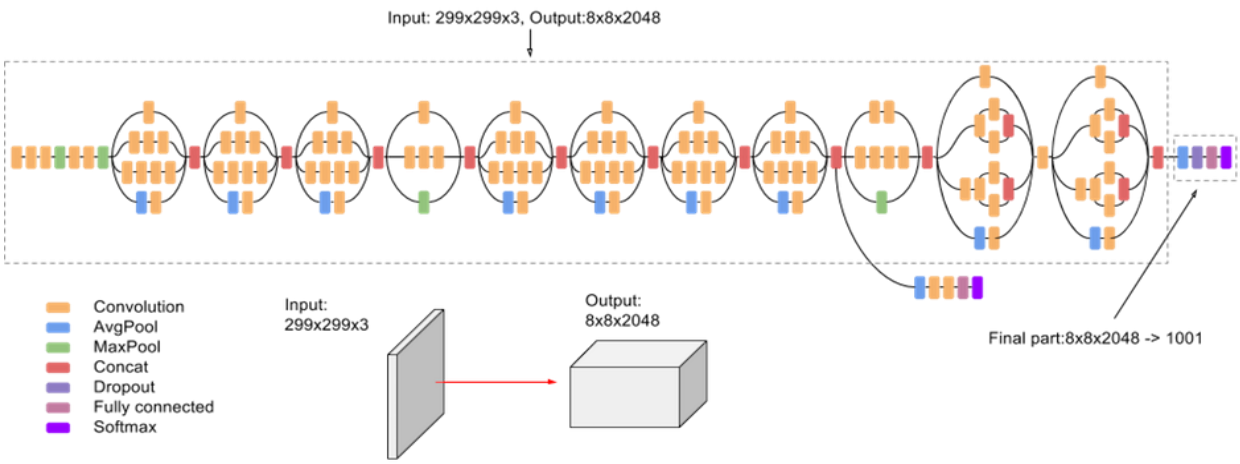

Fig. 3. Inception V3 Architecture [10].

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- Average Pooling Layer: Average pooling layer reduces the variance and complexity in the data. It also divides the input into rectangular pooling regions and computing the average values of each matrix to downsample the input features [10].
- Concat Layer: The Concat layer concatenates its multiple input blobs to one single output blob [10].
- Dropout Layer: The dropout layer randomly drops elements from a layer in the neural network. Dropout is a technique used to improve over-fit on neural networks [10].
- Fully Connected Layer: The fully connected (FC) layer in the CNN represents the feature vector for the input. This feature vector holds information that is vital to the input [10].
- Softmax Layer: The softmax assigns decimal probabilities to each class in a multi-class recognition problem. Those decimal probabilities must add up to 1.0. This additional constraint make training to converge more quickly [10].

D. *Clasificaation*

We trained our dataset of SGD [13] optimizer with initial learning rate of 0.01 and 0.9 momentum. We used a learning scheduler to set learning rate to 0.002 after 15 epochs and 0.0004 after 28 epochs.

III. Experimental Results

This section describes the results found in our experiments.

A. *Evaluation of model*

Our dataset was divided into three parts: training, validation and evaluation. We used training and validation parts of the dataset while training the model and we used evaluation part of our dataset during the evaluation of our model. We resized the images of evaluation part in 299 x x299 x 3We evaluated the accuracy of the model by true positive (TP), true negative (TN), false positive (FP) and false negative (FN) after classification.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

B. *Obtained Results*

The results obtained by running different models with food-11 dataset are given in Table 1.

TABLE I. Test Results

Model	*Accuracy*
CNN1[9]	82.23%
CNN2[9]	80.12%
Proposed Approach	92.86%

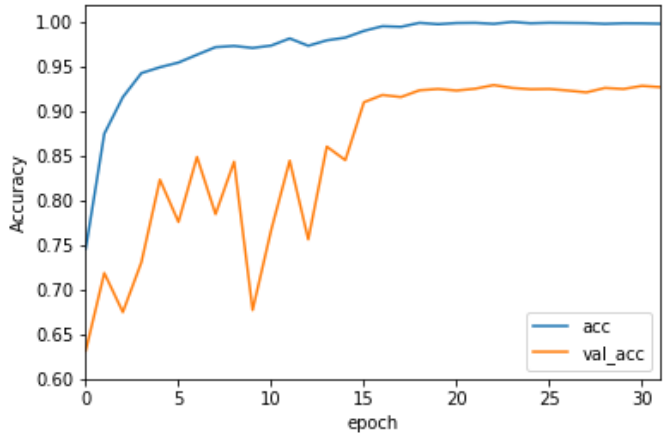

Fig. 4. Plot of model accuracy on training and validation datasets.

We applied the models described in [9]. We see that Fine tuned Alexnet [9] has 82.23% and Caffenet [9] has 80.12 % of accuracy with our dataset. The proposed approach which uses transfer learning technique with inception V3 convolution neural network has an accuracy of 92.86%. The transfer learning uses the knowledge earned from previous learning in new training dataset to classify images that is why our proposed approach has a better accuracy.

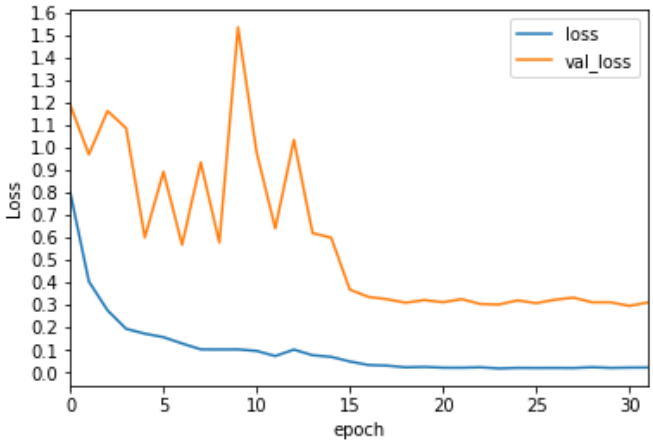

Fig. 5. Plot of model loss on training and validation datasets.

Though the accuracy is good from the accuracy curve in figure 3 we see that there is some gap between training and validation accuracy. This indicates that there was some overfitting but the gap gradually got smaller after some epochs. From loss curve in figure 4 we see that the loss 1.2 to 0.3.

IV. Conclusion and Future Work

In this paper we tried to use convolution neural network to classify food images. The convolutional neural networks use a variation of multilayer perceptrons and require minimal pre-processing. From our experiment we see that convolution

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neural network works very well in food classification task. However the there can be still some improvement to reduce the gap between training and test accuracy in our model. Other neural network models such as recurrent neural network, dilated convolutional neural network, etc can be applied to classify food images. Convolution neural network models take time for computation but once the model is trained it can be easily used for classification. We can also improve our research by applying feature based models which will take less computational time. We can also apply the different machine learning algorithms with larger datasets with more categories.

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